Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	**TE-AI&DS**	**Semester:**	**V**
Course Code:	**2015111**	**Course Name:**	**Statistics for Machine Learning and Data Science Lab**

Name of Student:	Kunal Yadav
Roll No. :	64
Experiment No.:	**8**
Title of the Experiment:	**Apply Clustering and Dimensionality Reduction Techniques for Pattern Discovery**
Date of Performance:	**22/09/26**
Date of Submission:	**29/09/26**

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :

Signature :

Date :

Aim: To apply clustering and dimensionality reduction techniques to the Pima Indians Diabetes Dataset to discover hidden patterns and visualize relationships among patients.

Objectives

After completing this experiment, the student will be able to:

1. Apply clustering and dimensionality reduction techniques to discover patterns in a high-dimensional dataset.
2. Interpret the resulting clusters and reduced-dimensional representations using suitable evaluation measures and visualizations.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains medical information for **768 female patients.** The Outcome variable indicates diabetes status.

The Outcome variable may be excluded during unsupervised learning and used later to compare the discovered clusters with the known diabetes categories.

Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select relevant numerical features for unsupervised analysis.
3. Handle invalid or missing values.
4. Standardize the selected features.
5. Apply K-Means clustering.
6. Select a suitable number of clusters using the Elbow Method.
7. Evaluate the clustering using the Silhouette Score.
8. Apply PCA to reduce the dimensionality of the dataset.
9. Visualize the observations in the reduced two-dimensional space.
10. Analyze the discovered clusters and interpret the results.

Description of the Experiment

Unsupervised learning identifies patterns in data without using a target variable during model development.

The experiment follows: High-Dimensional Dataset, Data Preprocessing, Feature Standardization, Clustering, Cluster Evaluation, Dimensionality Reduction, Visualization and Interpretation

Clustering groups similar observations, while dimensionality reduction represents high-dimensional data using fewer dimensions for visualization and analysis.

Detailed Description of Statistical Methods

K-Means Clustering

K-Means divides observations into *k*clusters by minimizing the distance between observations and their assigned cluster centroid.

The objective function is:

$$J = \sum_{j=1}^{k} \sum_{x_i \in C_j} \| x_i - \mu_j \|^2$$

where:

- C_j = cluster j
- μ_j = centroid of cluster j

Elbow Method

The Elbow Method evaluates different values of k using within-cluster sum of squares.

The value of k is selected by identifying the point where adding more clusters provides diminishing improvement.

Silhouette Score

The Silhouette Score evaluates how well an observation fits within its assigned cluster.

$$s = \frac{b-a}{max(a,b)}$$

where:

- a = average distance within the same cluster
- b = average distance to the nearest other cluster

A higher score generally indicates better-defined clusters.

Principal Component Analysis

PCA transforms correlated variables into a smaller number of uncorrelated principal components.

The principal components are obtained from the covariance matrix using eigenvalues and eigenvectors.

The first principal components capture the maximum possible variance in the data.

PCA can be used to represent the dataset in two dimensions for visualization.

Code:-

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.decomposition import PCA

import warnings
warnings.filterwarnings("ignore")
try:
    df = pd.read_csv("diabetes.csv")
except FileNotFoundError:
    from google.colab import files
    uploaded = files.upload()
    df = pd.read_csv("diabetes.csv")

print("=" * 70)
print("PIMA INDIANS DIABETES DATASET")
print("=" * 70)
```

```
print("\nDataset Shape:", df.shape)

print("\nFirst 5 Rows:")
print(df.head())

print("\nDataset Information:")
print(df.info())

print("\nStatistical Summary:")
print(df.describe())
print("\nMissing Values:")
print(df.isnull().sum())
invalid_columns = [
    "Glucose",
    "BloodPressure",
    "SkinThickness",
    "Insulin",
    "BMI"
]

df_clean = df.copy()

print("\nInvalid Zero Values:")
for col in invalid_columns:
    print(col, ":", (df_clean[col] == 0).sum())

# Replace zeros with NaN
```

```
for col in invalid_columns:
    df_clean[col] = df_clean[col].replace(0, np.nan)

# Replace missing values with median
for col in invalid_columns:
    df_clean[col] = df_clean[col].fillna(df_clean[col].median())

print("\nMissing Values After Imputation:")
print(df_clean.isnull().sum())
features = [
    "Pregnancies",
    "Glucose",
    "BloodPressure",
    "SkinThickness",
    "Insulin",
    "BMI",
    "DiabetesPedigreeFunction",
    "Age"
]

X = df_clean[features]

print("\nFeatures Used for Clustering:")
print(features)
# ------------------------------
# 6. STANDARDIZE FEATURES
# ------------------------------
```

```
scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

print("\nStandardization Completed.")

X_scaled_df = pd.DataFrame(
    X_scaled,
    columns=features
)

print("\nStandardized Data:")
print(X_scaled_df.head())

inertias = []
k_values = range(2, 11)

for k in k_values:

    kmeans = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=10
    )

    kmeans.fit(X_scaled)
```

```
    inertias.append(kmeans.inertia_)


# Plot Elbow Curve
plt.figure(figsize=(8, 5))

plt.plot(
    k_values,
    inertias,
    marker="o",
    color="blue",
    linewidth=2
)

plt.xlabel("Number of Clusters (k)")
plt.ylabel("Inertia / WCSS")
plt.title("Elbow Method")

plt.xticks(k_values)
plt.grid(True, alpha=0.3)

plt.show()

silhouette_scores = []

for k in range(2, 11):
```

```
    kmeans = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=10
    )

    labels = kmeans.fit_predict(X_scaled)

    score = silhouette_score(
        X_scaled,
        labels
    )

    silhouette_scores.append(score)
silhouette_results = pd.DataFrame({
    "Number of Clusters": range(2, 11),
    "Silhouette Score": silhouette_scores
})

print("\nSilhouette Scores:")
print(silhouette_results)



# Plot Silhouette Scores
plt.figure(figsize=(8, 5))

plt.plot(
```

```
    silhouette_results["Number of Clusters"],
    silhouette_results["Silhouette Score"],
    marker="o",
    color="green",
    linewidth=2
)

plt.xlabel("Number of Clusters (k)")
plt.ylabel("Silhouette Score")
plt.title("Silhouette Score for Different k Values")

plt.grid(True, alpha=0.3)

plt.show()
best_k = int(
    silhouette_results.loc[
        silhouette_results["Silhouette Score"].idxmax(),
        "Number of Clusters"
    ]
)

best_score = silhouette_results["Silhouette Score"].max()

print("\nBest Number of Clusters:", best_k)
print("Best Silhouette Score:", round(best_score, 4))

kmeans = KMeans(
```

```
    n_clusters=best_k,
    random_state=42,
    n_init=10
)

clusters = kmeans.fit_predict(X_scaled)

df_clean["Cluster"] = clusters

print("\nK-Means Clustering Completed.")

print("\nCluster Sizes:")
print(
    df_clean["Cluster"]
    .value_counts()
    .sort_index()
)

cluster_summary = df_clean.groupby("Cluster")[features].mean()

print("\nMean Feature Values for Each Cluster:")
print(cluster_summary)

plt.figure(figsize=(12, 6))

sns.heatmap(
    cluster_summary,
```

```
    annot=True,
    fmt=".2f",
    cmap="coolwarm"
)

plt.title("Mean Feature Values by Cluster")
plt.xlabel("Features")
plt.ylabel("Cluster")

plt.tight_layout()
plt.show()
cluster_centers = pd.DataFrame(
    kmeans.cluster_centers_,
    columns=features
)

print("\nStandardized Cluster Centers:")
print(cluster_centers)



plt.figure(figsize=(12, 6))

sns.heatmap(
    cluster_centers,
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
```

```
    center=0
)

plt.title("Standardized Cluster Centers")
plt.xlabel("Features")
plt.ylabel("Cluster")

plt.tight_layout()
plt.show()

cluster_outcome = pd.crosstab(
    df_clean["Cluster"],
    df_clean["Outcome"]
)

print("\nCluster vs Outcome:")
print(cluster_outcome)

cluster_outcome_percentage = pd.crosstab(
    df_clean["Cluster"],
    df_clean["Outcome"],
    normalize="index"
) * 100

print("\nCluster vs Outcome Percentage:")
print(cluster_outcome_percentage.round(2))
cluster_outcome_percentage.plot(
```

```
    kind="bar",
    figsize=(8, 5),
    color=["steelblue", "tomato"]
)

plt.xlabel("Cluster")
plt.ylabel("Percentage")
plt.title("Diabetes Outcome Distribution Within Clusters")

plt.legend(
    title="Outcome",
    labels=["No Diabetes", "Diabetes"]
)

plt.xticks(rotation=0)
plt.tight_layout()

plt.show()
pca = PCA(n_components=2)

X_pca = pca.fit_transform(X_scaled)

pca_df = pd.DataFrame(
    X_pca,
    columns=["PC1", "PC2"]
)
```

```
pca_df["Cluster"] = clusters
pca_df["Outcome"] = df_clean["Outcome"].values
explained_variance = pca.explained_variance_ratio_

print("\nPCA Explained Variance:")

print(
    "PC1:",
    round(explained_variance[0] * 100, 2),
    "%"
)

print(
    "PC2:",
    round(explained_variance[1] * 100, 2),
    "%"
)

print(
    "Total:",
    round(explained_variance.sum() * 100, 2),
    "%"
)
plt.figure(figsize=(7, 5))

plt.bar(
    ["PC1", "PC2"],
```

```
    explained_variance * 100,
    color=["steelblue", "orange"]
)

plt.ylabel("Explained Variance (%)")
plt.title("PCA Explained Variance")

plt.show()

plt.figure(figsize=(10, 7))

sns.scatterplot(
    data=pca_df,
    x="PC1",
    y="PC2",
    hue="Cluster",
    palette="Set1",
    s=70,
    alpha=0.75
)

plt.title("K-Means Clusters in PCA 2D Space")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")

plt.legend(title="Cluster")
```

```
plt.grid(True, alpha=0.2)
plt.tight_layout()

plt.show()
plt.figure(figsize=(10, 7))

sns.scatterplot(
    data=pca_df,
    x="PC1",
    y="PC2",
    hue="Outcome",
    palette={
        0: "steelblue",
        1: "red"
    },
    s=70,
    alpha=0.75
)

plt.title("PCA Representation Colored by Diabetes Outcome")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")

plt.legend(
    title="Outcome",
    labels=["No Diabetes", "Diabetes"]
)
```

```
plt.grid(True, alpha=0.2)
plt.tight_layout()

plt.show()

loadings = pd.DataFrame(
    pca.components_.T,
    columns=["PC1", "PC2"],
    index=features
)

print("\nPCA Feature Loadings:")
print(loadings)



plt.figure(figsize=(8, 6))

sns.heatmap(
    loadings,
    annot=True,
    cmap="coolwarm",
    center=0
)

plt.title("PCA Feature Loadings")
```

```
plt.tight_layout()
plt.show()

print("\n")
print("=" * 70)
print("FINAL RESULTS")
print("=" * 70)

print("\nOriginal Dataset Shape:", df.shape)

print("\nNumber of Features Used:", len(features))

print("\nSelected Number of Clusters:", best_k)

print(
    "Silhouette Score:",
    round(best_score, 4)
)

print("\nPCA Variance Explained:")

print(
    "PC1:",
    round(explained_variance[0] * 100, 2),
    "%"
)
```

```
print(
    "PC2:",
    round(explained_variance[1] * 100, 2),
    "%"
)

print(
    "Total:",
    round(explained_variance.sum() * 100, 2),
    "%"
)

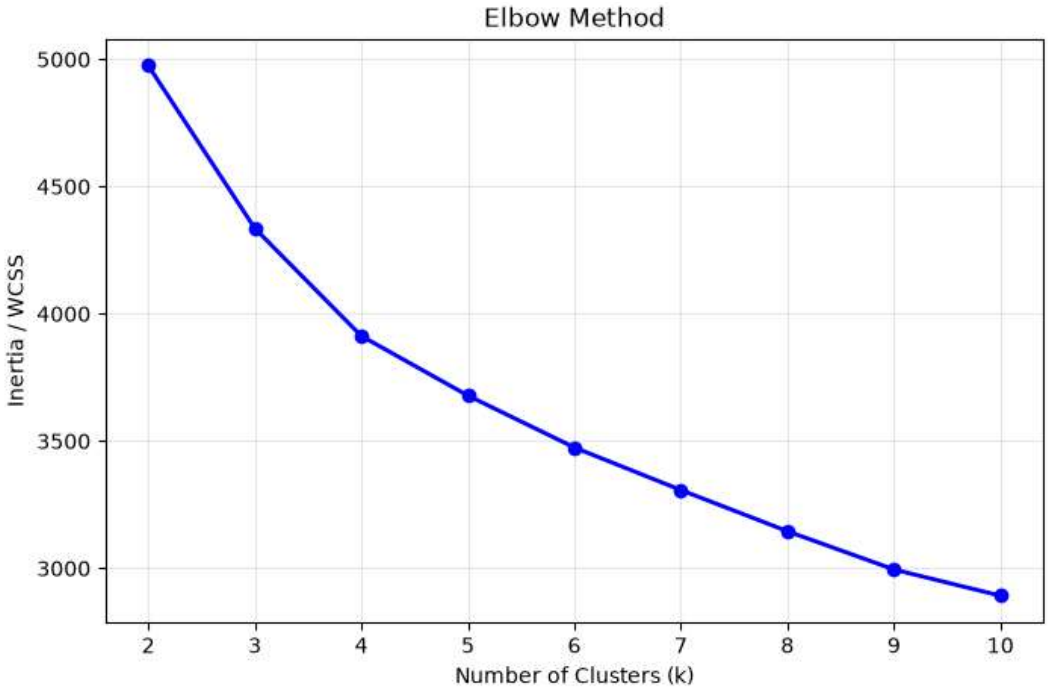
print("\nCluster Sizes:")
print(
    df_clean["Cluster"]
    .value_counts()
    .sort_index()
)
print("\nCluster Mean Values:")
print(cluster_summary.round(2))

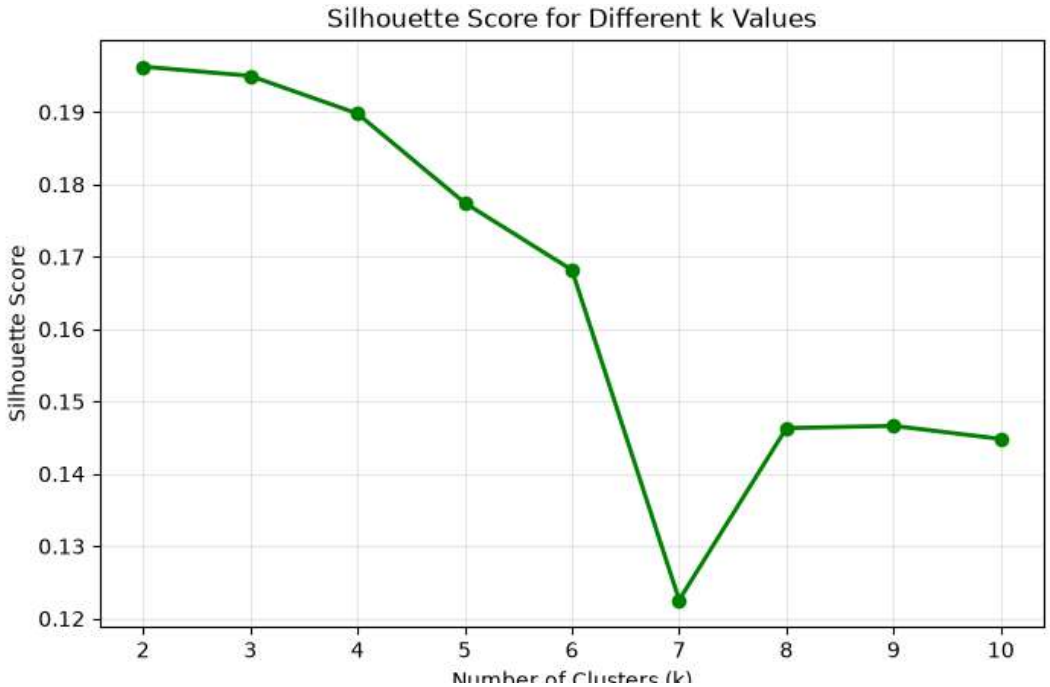
print("\nCluster vs Outcome:")
print(cluster_outcome)

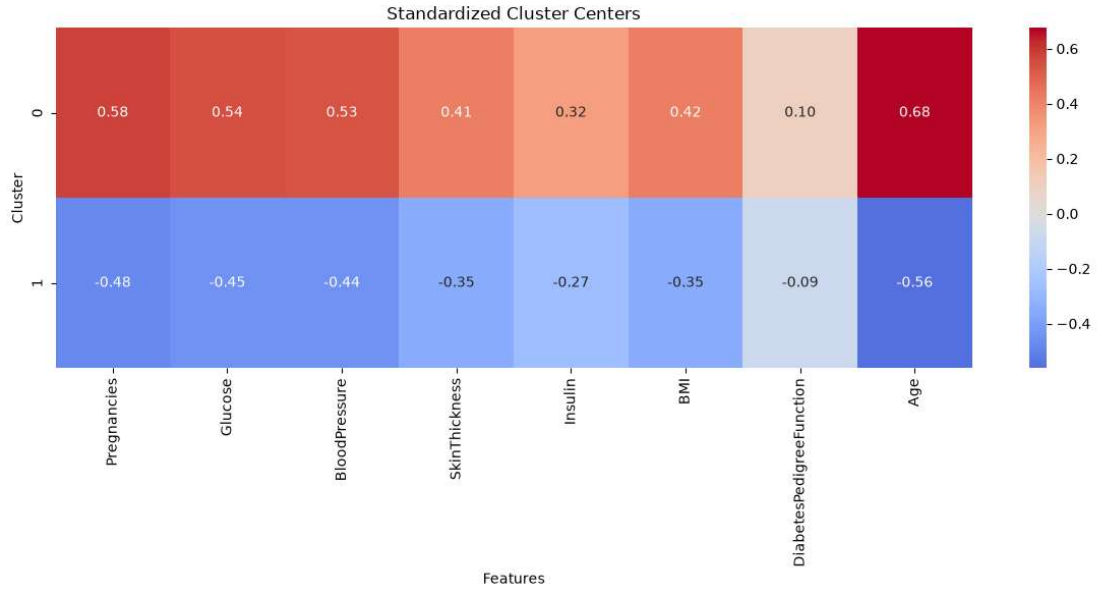
print("\nExperiment Completed Successfully!")

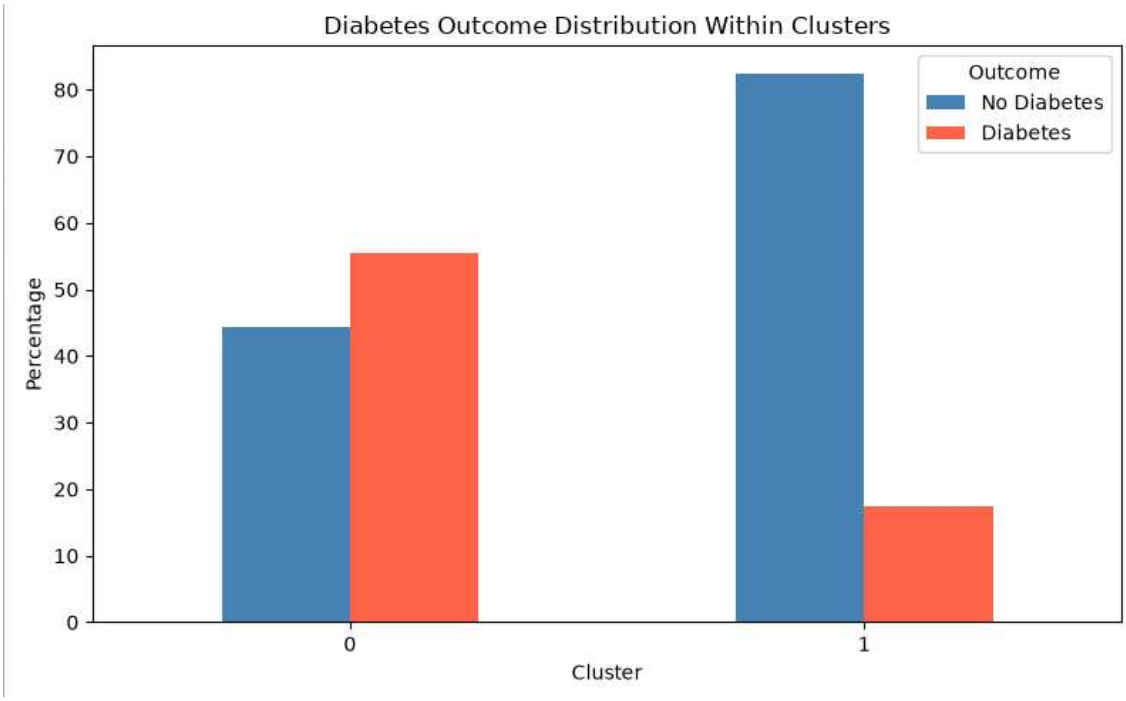
print("=" * 70)
```

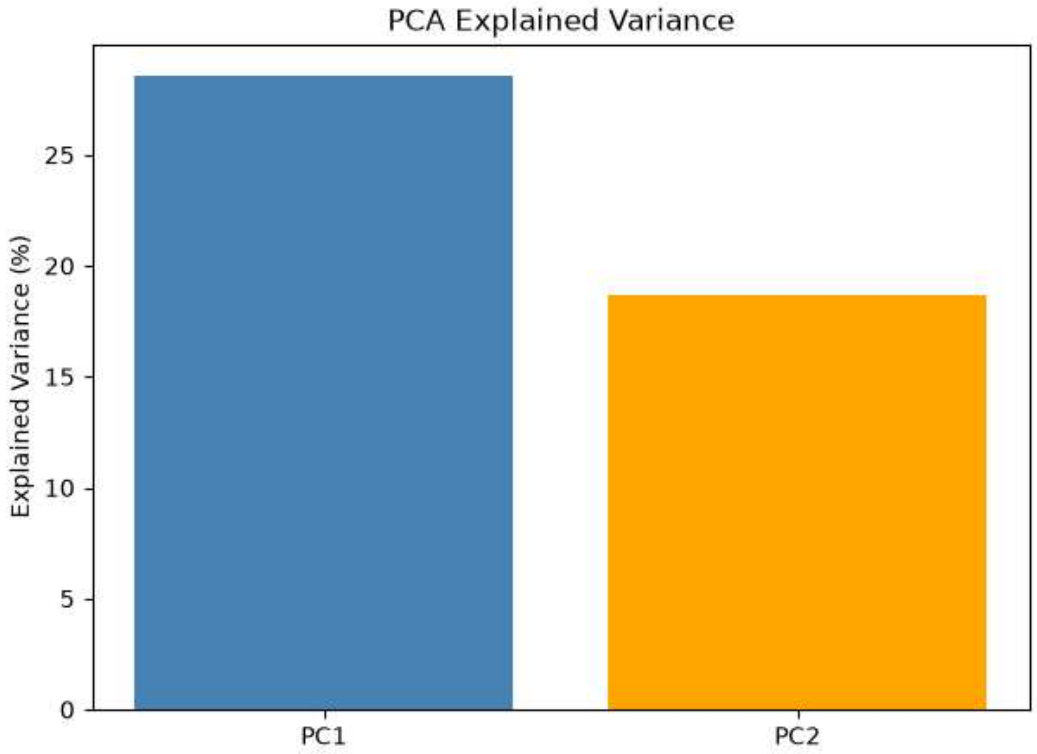
Output:-

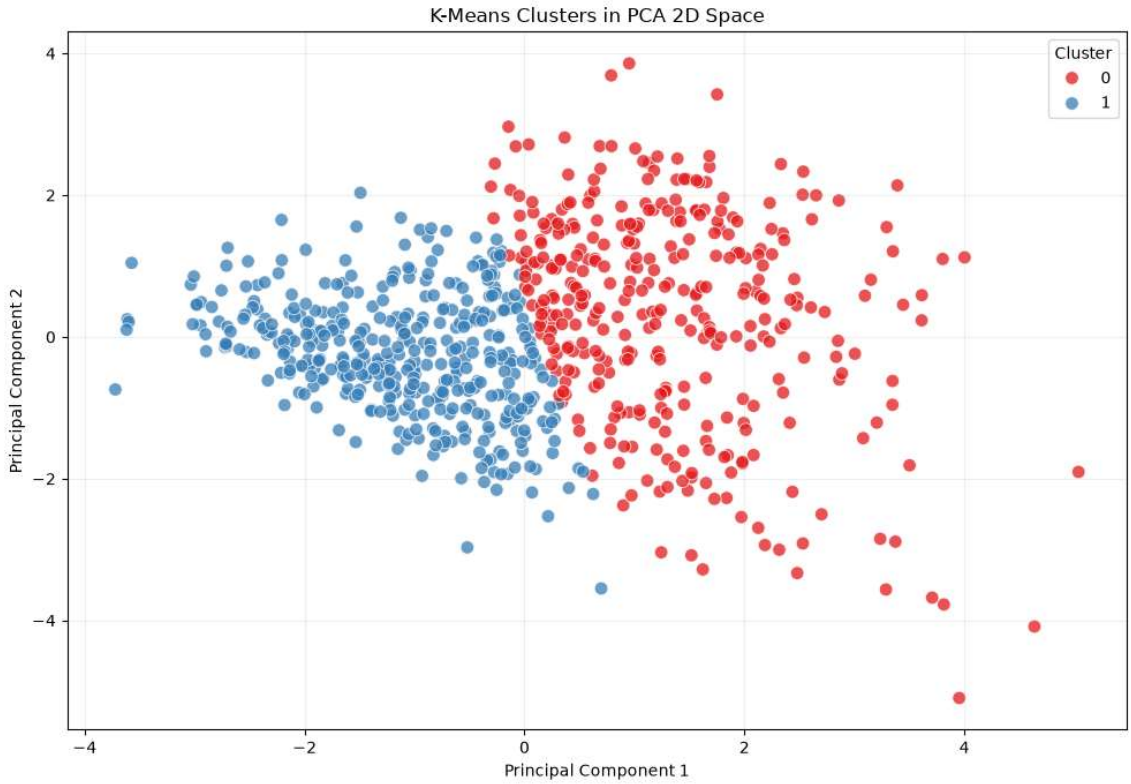

Elbow Method
Inertia / WCSS
Number of Clusters (k)
5000
4500
4000
3500
3000
2
3
4
5
6
7
8
9
10

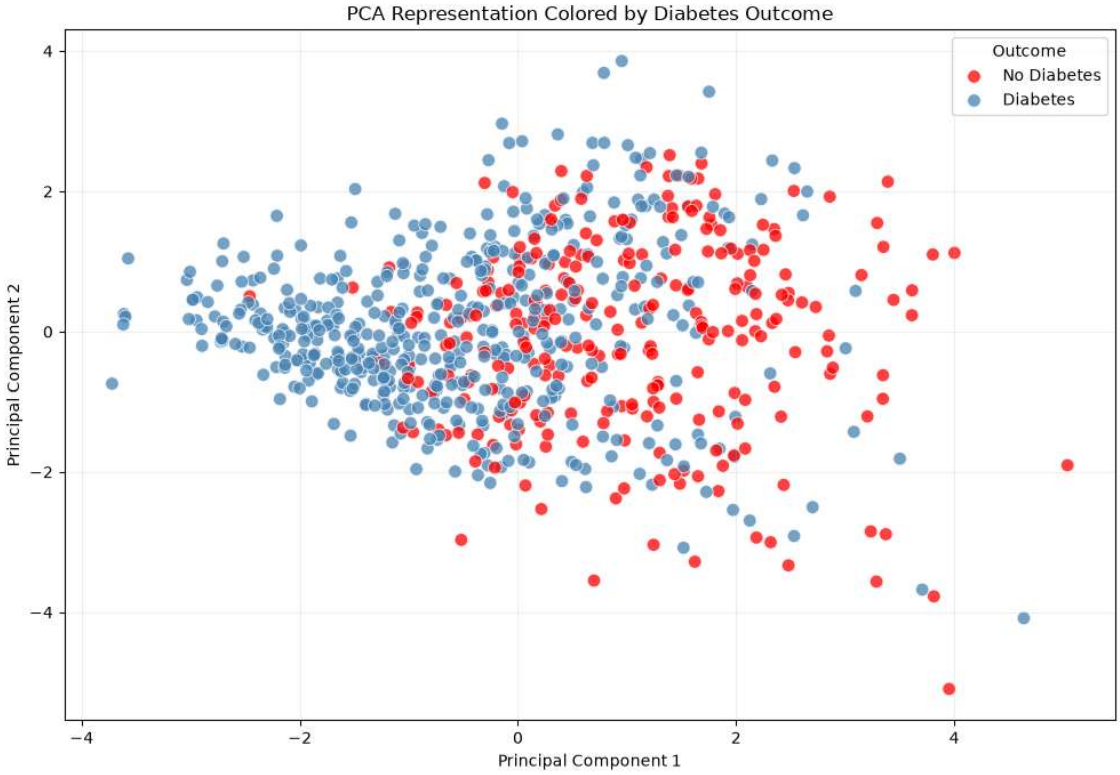

Silhouette Score for Different k Values
Silhouette Score
0.19
0.18
0.17
0.16
0.15
0.14
0.13
0.12
2
3
4
5
6
7
8
9
10

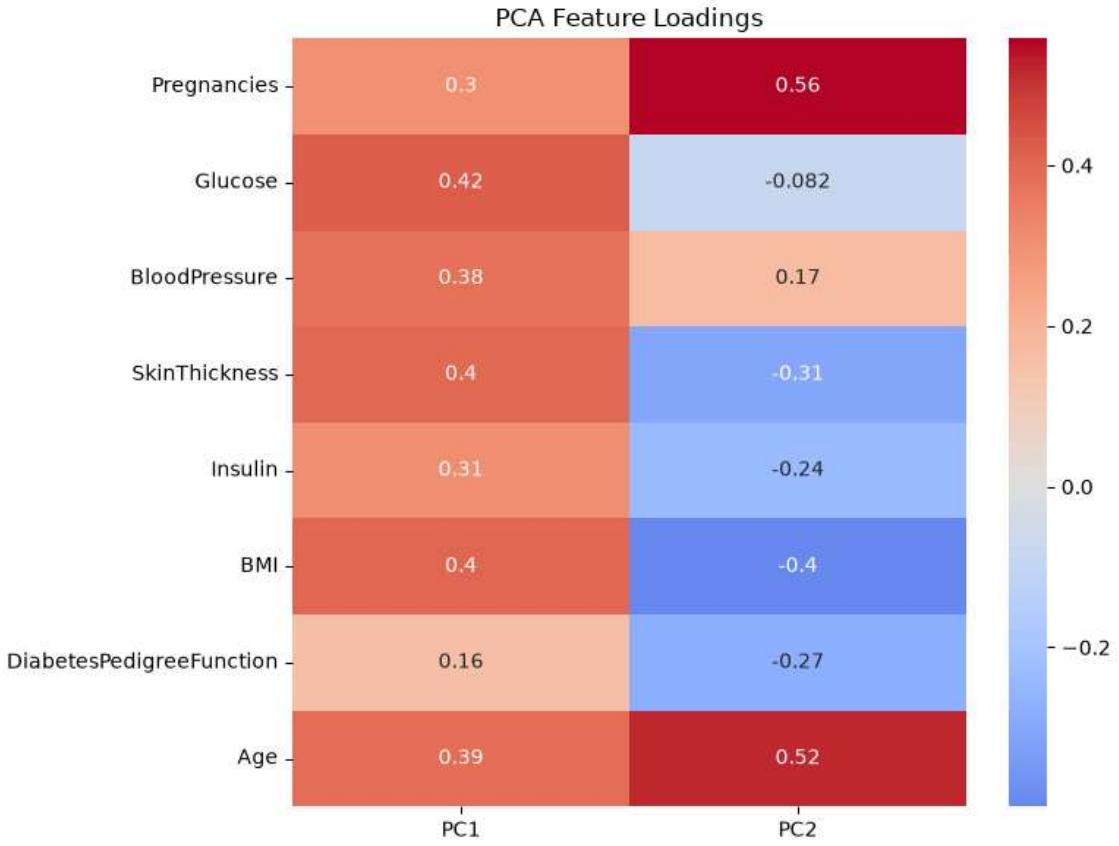

Standardized Cluster Centers
Cluster
0
1
0.58
0.54
0.53
0.41
0.32
0.42
0.10
0.68
-0.48
-0.45
-0.44
-0.35
-0.27
-0.35
-0.09
-0.56
0.6
0.4
0.2
0.0
−0.2
−0.4
Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Features


Diabetes Outcome Distribution Within Clusters
Outcome
No Diabetes
Diabetes
Percentage
80
70
60
50
40
30
20
10
0
0
1
Cluster


PCA Explained Variance
Explained Variance (%)
0
5
10
15
20
25
PC1
PC2


K-Means Clusters in PCA 2D Space
Cluster
0
1
Principal Component 2
Principal Component 1
4
2
0
−2
−4
−4
−2
0
2
4


PCA Representation Colored by Diabetes Outcome
Outcome
No Diabetes
Diabetes
Principal Component 2
Principal Component 1
4
2
0
−2
−4
−4
−2
0
2
4




```
======================================================================
PIMA INDIANS DIABETES DATASET
======================================================================

Dataset Shape: (768, 9)

First 5 Rows:
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age  Outcome
0            6      148             72             35        0  33.6                     0.627   50        1
1            1       85             66             29        0  26.6                     0.351   31        0
2            8      183             64              0        0  23.3                     0.672   32        1
3            1       89             66             23       94  28.1                     0.167   21        0
4            0      137             40             35      168  43.1                     2.288   33        1

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
```

```
Statistical Summary:
       Pregnancies     Glucose  BloodPressure  ...  DiabetesPedigreeFunction         Age     Outcome
count   768.000000  768.000000     768.000000  ...                768.000000  768.000000  768.000000
mean      3.845052  120.894531      69.105469  ...                  0.471876   33.240885    0.348958
std       3.369578   31.972618      19.355807  ...                  0.331329   11.760232    0.476951
min       0.000000    0.000000       0.000000  ...                  0.078000   21.000000    0.000000
25%       1.000000   99.000000      62.000000  ...                  0.243750   24.000000    0.000000
50%       3.000000  117.000000      72.000000  ...                  0.372500   29.000000    0.000000
75%       6.000000  140.250000      80.000000  ...                  0.626250   41.000000    1.000000
max      17.000000  199.000000     122.000000  ...                  2.420000   81.000000    1.000000

[8 rows x 9 columns]

Missing Values:
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

Invalid Zero Values:
Glucose : 5
BloodPressure : 35
SkinThickness : 227
Insulin : 374
BMI : 11

Missing Values After Imputation:
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64
```

```
Features Used for Clustering:
['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI', 'DiabetesPedigreeFunction', 'Age']

Standardization Completed.

Standardized Data:
   Pregnancies   Glucose  BloodPressure  SkinThickness   Insulin       BMI  DiabetesPedigreeFunction       Age
0     0.639947  0.866045      -0.031990       0.670643 -0.181541  0.166619                  0.468492  1.425995
1    -0.844885 -1.205066      -0.528319      -0.012301 -0.181541 -0.852200                 -0.365061 -0.190672
2     1.233880  2.016662      -0.693761      -0.012301 -0.181541 -1.332500                  0.604397 -0.105584
3    -0.844885 -1.073567      -0.528319      -0.695245 -0.540642 -0.633881                 -0.920763 -1.041549
4    -1.141852  0.504422      -2.679076       0.670643  0.316566  1.549303                  5.484909 -0.020496

Silhouette Scores:
   Number of Clusters  Silhouette Score
0                   2          0.196279
1                   3          0.195004
2                   4          0.189791
3                   5          0.177453
4                   6          0.168170
5                   7          0.122561
6                   8          0.146312
7                   9          0.146626
8                  10          0.144814

Best Number of Clusters: 2
Best Silhouette Score: 0.1963
```

```
K-Means Clustering Completed.

Cluster Sizes:
Cluster
0    351
1    417
Name: count, dtype: int64

Mean Feature Values for Each Cluster:
         Pregnancies     Glucose  BloodPressure  ...        BMI  DiabetesPedigreeFunction        Age
Cluster                                          ...
0           5.777778  137.914530      78.829060  ...  35.324501                  0.506145  41.116809
1           2.218225  107.971223      66.964029  ...  30.040048                  0.443031  26.611511

[2 rows x 8 columns]

Standardized Cluster Centers:
   Pregnancies   Glucose  BloodPressure  SkinThickness   Insulin       BMI  DiabetesPedigreeFunction       Age
0     0.576129  0.541724       0.532839       0.414946  0.322341  0.418759                  0.102876  0.675322
1    -0.479879 -0.451221      -0.443820      -0.345624 -0.268490 -0.348800                 -0.085689 -0.562499

Cluster vs Outcome:
Outcome    0    1
Cluster
0        156  195
1        344   73

Cluster vs Outcome Percentage:
Outcome      0      1
Cluster
0        44.44  55.56
1        82.49  17.51

PCA Explained Variance:
PC1: 28.54 %
PC2: 18.69 %
Total: 47.23 %

PCA Feature Loadings:
                              PC1       PC2
Pregnancies              0.301434  0.557960
Glucose                  0.423980 -0.082050
BloodPressure            0.377177  0.171980
SkinThickness            0.397103 -0.308746
Insulin                  0.307134 -0.235780
BMI                      0.401844 -0.398489
DiabetesPedigreeFunction 0.156874 -0.273482
Age                      0.386013  0.517560
```

```
======================================================================
FINAL RESULTS
======================================================================

Original Dataset Shape: (768, 9)

Number of Features Used: 8

Selected Number of Clusters: 2
Silhouette Score: 0.1963

PCA Variance Explained:
PC1: 28.54 %
PC2: 18.69 %
Total: 47.23 %

Cluster Sizes:
Cluster
0    351
1    417
Name: count, dtype: int64

Cluster Mean Values:
         Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin    BMI  DiabetesPedigreeFunction    Age
Cluster
0               5.78   137.91          78.83          32.74   168.49  35.32                      0.51  41.12
1               2.22   107.97          66.96          26.05   117.25  30.04                      0.44  26.61

Cluster vs Outcome:
Outcome    0    1
Cluster
0        156  195
1        344   73

Experiment Completed Successfully!
======================================================================
```

Conclusion

Clustering and dimensionality reduction techniques were applied to the Pima Indians Diabetes Dataset. K-Means clustering was used to group patients based on similarities in their medical attributes, while PCA was used to reduce the dimensionality of the dataset for visualization. The resulting clusters were evaluated and interpreted using appropriate measures and visualizations.

Questions to be Answered by Students

1. What value of k was selected using the Elbow Method, and why?

The selected value was **k = 2**. The Elbow Method is used to identify the point where increasing the number of clusters gives diminishing improvement in within-cluster variation. The Silhouette analysis also supported **2 clusters**, which was the value with the highest Silhouette Score.

2. What Silhouette Score was obtained, and what does it indicate about the quality of the clusters?

The Silhouette Score obtained was **0.1963** for **k = 2**.

A score closer to 1 indicates well-separated and compact clusters, while a score near 0 indicates overlapping clusters. Therefore, the score of 0.1963 indicates relatively weak separation and considerable overlap between the two clusters. The clusters exist as groups in the feature space, but they are not strongly separated.

3. Which variables contributed most to the first two principal components?

From your PCA loadings:

PC1 was mainly influenced by:

- Glucose = **0.424**
- BMI = **0.402**
- SkinThickness = **0.397**
- Age = **0.386**
- BloodPressure = **0.377**

PC2 was mainly influenced by:

- Pregnancies = **0.558**
- Age = **0.518**
- BMI = **-0.398**
- SkinThickness = **-0.309**

Therefore, Glucose, BMI, SkinThickness, BloodPressure and Age were important contributors to PC1, while Pregnancies and Age were particularly important for PC2.

4. What patterns were observed in the PCA visualization of the clusters?

The PCA representation showed the observations distributed into two groups corresponding to the K-Means clusters. However, because the Silhouette Score was only **0.1963**, the clusters are expected to have **substantial overlap** rather than forming two completely separate regions.

The first two principal components explained **47.23%** of the total variance:

- PC1 = **28.54%**
- PC2 = **18.69%**

Thus, the PCA plot provides a useful two-dimensional visualization, but it does not represent all the variation present in the original eight-dimensional dataset.

5. How do the discovered clusters compare with the known Outcome categories?

The comparison was:

Cluster	Outcome = 0	Outcome = 1
Cluster 0	156	195
Cluster 1	344	73

In percentage terms:

Cluster	Outcome = 0	Outcome = 1
Cluster 0	44.44%	**55.56%**
Cluster 1	**82.49%**	17.51%

Thus, **Cluster 0 contained a higher proportion of patients with Outcome = 1**, whereas **Cluster 1 contained a much higher proportion of patients with Outcome = 0**. This shows an association between the clusters discovered without using `Outcome` and the known diabetes categories.